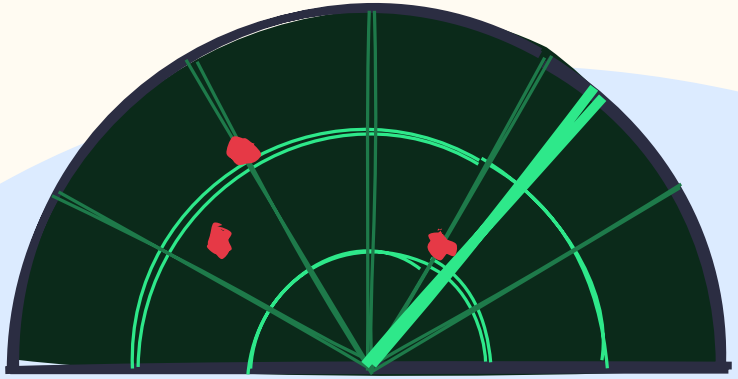
Young Innovators · Class 18

Processing + Arduino: Live Sensor Radar

Your Arduino senses. Your laptop draws.

Echo | Serial | cos & sin



Did you know?
Airports track every plane with radar, sweeping a beam round and plotting echoes, just like you will today.

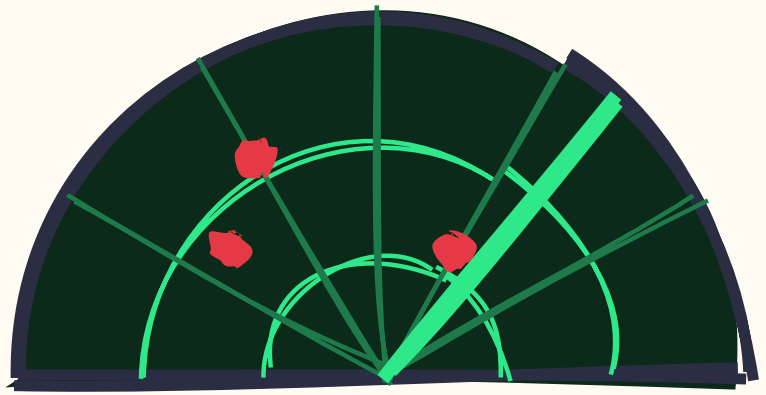
LEARN

Let's think...

How do ships 'see' in thick fog?

Fog at the harbour

Karachi port, thick fog, zero visibility. Yet the captain's screen shows every boat as a glowing dot. No camera can see through fog. What is drawing those dots?



A drone?

GPS?

Radar echoes!

Today: you build a **sonar radar**: a servo sweeps an ultrasonic sensor and Processing draws what it finds, live.

LEARN

Learning objectives

By the end of this class I can...

1	**Echo distance:** calculate distance from an ultrasonic **echo time** and sweep it with a **servo**	**Proof:** my Serial Monitor shows angle and distance pairs
2	**Serial link:** send data from Arduino to the laptop as **angle,distance** lines over **Serial**	**Proof:** Processing prints the numbers my Arduino sends
3	**Radar screen:** turn angle and distance into **x, y** with **cos** and **sin** and draw a live radar	**Proof:** objects appear as dots in the right place

LEARN

Objective 1 · Echo distance · video

Video time!



How does RADAR work? | James May Q&A

BBC Earth Science · about 6 minutes · youtu.be/ywDE57CtaTM

Watch for...

1. How does radar find an object's **distance**?
2. Why does the beam **sweep** round?
3. What can radar see that cameras can't?

Scan to watch

After the video: Explain to a partner how a radar knows both direction and distance.

Objective 1 · Echo distance

Echo time → distance

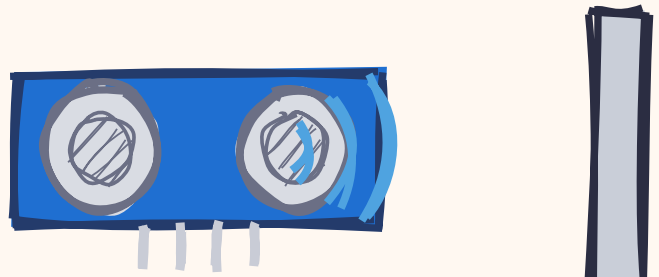
The idea

The HC-SR04 sends a 40 kHz **ping** and times the **echo**. Sound travels about **343 m/s = 0.0343 cm/µs**.

The sound goes there **and back**, so:
distance (cm) = time (µs) × 0.0343 ÷ 2

Echo of 1000 µs → 1000 × 0.0343 ÷ 2 ≈ **17 cm**. A servo turns the sensor 15° → 165° so we scan a whole arc.

Example: shouting in a valley



Shout across a valley and the echo comes back 2 seconds later. Sound went there and back, so the cliff is about 343 m away, not 686.

Objective 2 · Serial link

Serial: Arduino talks to the laptop

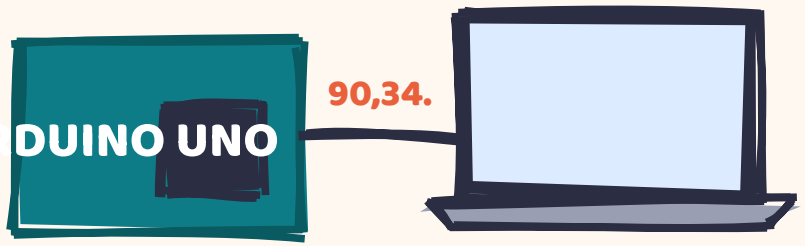
The idea

Serial.print() sends text down the USB cable. Processing can read it with the **Serial** library.

We send one line per reading: **angle,distance.** e.g. **90,34.**
The comma separates the two numbers. The full stop marks the end of the message.

Both sides must use the **same speed** (baud rate): 9600.

Example: a walkie-talkie code



Pilots say 'Roger. Over.' so the other side knows when a message ends. Our full stop is the 'over'.

Objective 3 · Radar screen

From angle + distance to x, y

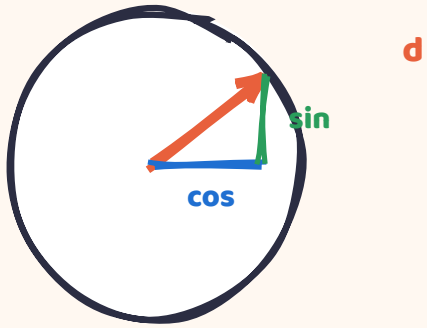
The idea

Our sensor gives **polar** data: an angle and a distance. The screen needs **x, y**.

x = cx + d × cos(angle)
y = cy − d × sin(angle)

Minus because screen y grows **downwards**. Processing angles are in radians: use **radians(angle)**.

Example: a clock hand



The tip of a clock hand: we know its length and its angle. cos tells you how far across it reaches, sin how far up.

Real world

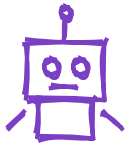
Sweeping sensors



Air traffic radar
Tracks planes around Islamabad airport



Parking sensors
Ultrasonic beeps get faster near a wall



Robot vacuums
Spinning LiDAR maps the whole room



Weather radar
Sees rain clouds coming

? Why do bats and dolphins use sound, but planes use radio waves?

LiDAR
Self-driving cars use LiDAR: the same idea with laser light, a million points per second.

Quick quiz

Radar maths

Question	Answer
1. Echo time 2000 µs. Distance?	
2. Why divide by 2?	
3. Arduino sends '45,20.' What is the angle?	
4. At angle 90°, cos = ? sin = ?	
5. Why y = cy − …, not + ?	

Answers

Cover! 1: ≈ 34 cm · 2: there and back · 3: 45° · 4: 0 and 1 · 5: screen y grows downwards

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human radar
2. Our radar kit (per group)
3. Arduino: sweep and send
4. Processing: the radar screen
5. Run, tune and test
6. Test like an engineer
7. Radar challenges

Class game

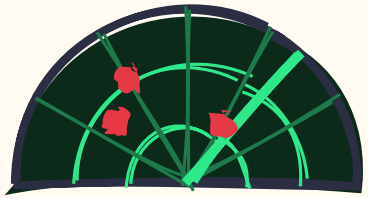
Human radar

How to play

1. One student is the **radar** in the middle, arm pointing out like a beam.
2. 5 students stand in the 'room' as objects.
3. Radar turns slowly. When the arm points at someone, they call their distance in steps.
4. A scribe plots each (angle, distance) on a paper semicircle. Does the map match the room?

Then discuss

What happens to an object hiding exactly behind another one?



Big idea

The radar only knows angle + distance. The drawing step turns it into a map.

MAKE

Build it

Our radar kit (per group)



Arduino Uno
Reads the sensor, drives the servo



HC-SR04
Ultrasonic distance sensor



SG90 servo
Sweeps 15° to 165°



Processing 4
Draws the radar screen



Jumper wires
Plus a little hot glue / tape



Mount
Card bracket for the sensor

Wiring: Trig → 10, Echo → 11, servo signal → 12. Servo and sensor VCC → 5V, all GND together.

One app at a time
Close the Arduino Serial Monitor before running Processing: only one program can use the port.

MAKE

Objective 1 · Echo distance · Build step 1 of 3

Arduino: sweep and send

```
#include <Servo.h>
const int TRIG = 10, ECHO = 11;
Servo s;

int readCm() {
  digitalWrite(TRIG, LOW);  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG, LOW);
  long t = pulseIn(ECHO, HIGH, 30000);   // µs
  return t * 0.0343 / 2;
}

void sendReading(int a) {
  s.write(a); delay(30);
  Serial.print(a); Serial.print(",");
  Serial.print(readCm()); Serial.print(".");
}

void setup() {
  pinMode(TRIG, OUTPUT); pinMode(ECHO, INPUT);
  s.attach(12); Serial.begin(9600);
}

void loop() {
  for (int a = 15; a <= 165; a++) sendReading(a);
  for (int a = 165; a >= 15; a--) sendReading(a);
}
```

Upload

Upload, then open Serial Monitor (9600). You should see 15,40.16,41.17,39...

Timeout

pulseIn(..., 30000) gives up after 30 ms. 0 means 'nothing in range'.

Then close

Close Serial Monitor before Processing!

MAKE

Objective 3 · Radar screen · Build step 2 of 3

Processing: the radar screen

```
import processing.serial.*;
Serial port;  int ang = 0, dist = 0;

void setup() {
  size(900, 500);
  println(Serial.list());               // find your port
  port = new Serial(this, Serial.list()[0], 9600);
  port.bufferUntil('.');
  background(0);
}

void serialEvent(Serial p) {
  String msg = trim(p.readStringUntil('.'));
  msg = msg.substring(0, msg.length() - 1);  // drop '.'
  String[] v = split(msg, ',');
  if (v.length == 2) { ang = int(v[0]); dist = int(v[1]); }
}

void draw() {
  fill(0, 12); noStroke(); rect(0, 0, width, height); // fade
  float cx = width/2, cy = height - 20, a = radians(ang);
  stroke(46, 232, 138); strokeWeight(2);
  line(cx, cy, cx + 420*cos(a), cy - 420*sin(a));     // beam
  if (dist > 0 && dist < 40) {
    float r = map(dist, 0, 40, 0, 420);
    fill(255, 60, 60); noStroke();
    circle(cx + r*cos(a), cy - r*sin(a), 12);          // object
  }
}
```

Port

If it fails, look at the printed list and change [0] to the right number, e.g. [1].

Fade

A see-through black rect each frame makes old dots fade slowly, like real radar.

map()

map() scales 0 to 40 cm onto 0 to 420 pixels.

MAKE

Objective 3 · Build step 3 of 3

Run, tune and test

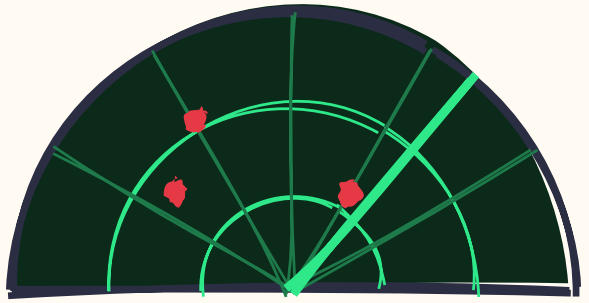
a Run the Arduino first (powered by USB), then press ▶ in Processing.

b Place a pencil case 20 cm away at about 45°. Does a red dot appear there?

c Move it to 90° and 135°. Does the dot follow?

d Add the arcs and labels: draw arcs at 10, 20, 30, 40 cm and text the angle.

e Change the range from 40 to 100 cm. What changes, and what gets noisier?



Check: Objects appear as dots at the right angle and distance.

Keep fingers out of the servo's sweep, and don't force the horn by hand.

Test like an engineer

Test like an engineer

Test	Result	Unit
Real 20 cm at 45°: radar shows		cm, °
Real 30 cm at 90°: radar shows		cm, °
Smallest object it can see		
Soft object (jumper): seen?		
Time for one full sweep		s

Debug checklist

Processing error 'Port busy'? Close Serial Monitor.
Dots in random places? Check both sides use 9600 and the '.' end marker.
Always 0? Check Trig/Echo pins.

Soft and angled surfaces scatter sound, so the echo never comes back. That's why stealth planes have angled shapes!

MAKE

Level up!

Radar challenges

Choose your challenge

- **Alarm:** beep a buzzer on the Arduino when something is closer than 15 cm.
- **Labels:** print the distance next to each dot in Processing.
- **Map mode:** keep dots forever (no fade) and map your desk.
- **Two-way:** click in Processing to send a command that stops the servo.

Think like an engineer

Why does a faster sweep make the radar less accurate?

Boss level

Make a 360° radar with two sensors back-to-back.

Objective check

Can I really do it? Prove it.

01	An echo takes 1500 µs. How far is the object? Show your working.	○ ○ ○
02	Show the Serial data your Arduino sends. What do the comma and full stop do?	○ ○ ○
03	Point to the line that uses cos and sin. Why the minus sign?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Radar demo: the teacher hides an object; can your radar find its angle and distance?

Show what you know

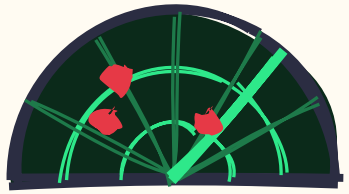
What did we learn?

- [x] distance = time × 0.0343 ÷ 2
- [x] A servo sweeps the sensor to scan an arc
- [x] Serial sends 'angle,distance.' lines to the laptop
- [x] x = d·cos(a), y = −d·sin(a) turns polar into screen
- [x] Arduino senses, Processing visualises

Exit ticket: why do we divide the echo time by 2?

At home

Find a car with parking sensors. Listen to how the beeps change.



Next class:
Python Pygame: Catch the Falling Stars!